

Ferroelectric Tunnel Junction (FTJ) Storage Controller Masterclass

A complete executive and engineering guide: Demystifying non-volatile memory physics, crossbar architecture, lock-free NVMe queues, and silicon IP licensing.

Quantum Tunneling (TER)

Merz's Law Dynamic Switching

Lock-Free NVMe Queues

Hamming SECDED (72, 64)

28nm Mature Foundry Strategy

Silicon IP Licensing

Why Modern AI Workloads Break 3D NAND SSDs

The fundamental mismatch between machine learning write patterns and flash physics.

1. Block-Based Immutability

Traditional 3D NAND flash **cannot overwrite single bytes** in place.

Data is written in 4KB/16KB pages, but can only be erased in giant **4MB to 16MB blocks** using high voltage (~20V).

NAND: 1 byte edit = 16 MB copy & block erase

2. Write Amplification (WAF)

Background **Garbage Collection (GC)** must constantly relocate stale blocks to free up erase space.

Writing 1 GB of random AI KV-cache data can trigger **3 to 10 GB of physical flash wear**.

Result: Latency spikes from 100 μ s to **3 ms** during erases.

3. Finite Endurance Death

Enterprise TLC/QLC flash withstands only **~1,000 to 3,000 write cycles** before the oxide breaks down permanently.

Continuous AI inferencing and vector database updates destroy flash drives in **months**.

Flash Endurance: ~3,000 P/E Cycles

Ferroelectric Tunnel Junction (FTJ) Physics

Quantum tunneling meets permanent electric polarization in nanoscale thin films.

How FTJ Actually Works

An FTJ cell consists of an ultra-thin (~1-2 nm) ferroelectric barrier (HZO) sandwiched between metal electrodes.

- **Quantum Mechanical Tunneling:** Electrons tunnel straight through the barrier without damaging it.
- **Tunnel Electroresistance (TER):** Applying a small voltage pulse flips the electric polarization:
 - Polarization UP: High Resistance State (HRS) = **0**
 - Polarization DOWN: Low Resistance State (LRS) = **1**
- **Non-Volatile:** The polarization remains frozen in place even after power is disconnected.

3D NAND Flash vs. FTJ Memory

Metric	3D NAND Flash	FTJ (This Project)
Access Granularity	Block (4MB - 16MB)	Byte-Level (1 Byte)
Write Latency	~100 μ s (+ 3ms GC)	~237 ns (dynamic)
Endurance	~3,000 Cycles	> 10 ¹⁰ Cycles
Write Amplification	3.0x - 10.0x	~1.0x (In-place write)
Block Erase Needed?	Yes (High Voltage 20V)	Zero Erases Ever

Crossbar Physical Modeling & Electrical Realities

Why our simulator is not a toy: Modeling real semiconductor electrical constraints.

1. Merz's Law

Ferroelectric switching speed is governed by electric field strength and temperature:

$$t_{sw} \propto \exp(\alpha / E_{applied})$$

Our simulator calculates dynamic delay in nanoseconds rather than fixed fake numbers.

At 85°C: switching settles at **237 ns**.

2. Wire IR-Drop

Microscopic tungsten and copper wires have electrical resistance ($V = I \times R$).

In our 512×512 crossbar grid, peripheral cells suffer a **443 mV voltage drop**.

The controller compensates pulse duration to guarantee reliable state switching.

3. HAR-SM Refresh

Half-Select Disturb: Writing cell (Row 5, Col 10) causes unselected cells along the same line to experience half-voltage.

HAR-SM: Hardware Autonomous Refresh State Machine monitors disturb counts and re-strengthens cells automatically.

Controller Architecture: From Verilog RTL to Windows

How commands travel through lock-free queues, ECC, and the operating system.

1. Lock-Free NVMe Queues

Standard OS mutex locks take microseconds—too slow for nanosecond storage.

We implemented **Lock-Free Ring Buffers** in C++20 and Verilog HDL using atomic memory fences.

Host SQ (Submission) and CQ (Completion) process in parallel with zero lock overhead.

2. Hamming SEDED (72, 64)

For every 64 bits of data, hardware generates 8 parity bits.

- **Single-Bit Error:** Corrected in hardware instantly in real-time (100% data recovery).
- **Double-Bit Error:** Detected immediately, alerting the system before silent data corruption occurs.

3. Live WinFSP Virtual Mount

Integrated with the **Windows File System Proxy (WinFSP)** driver.

The OS mounts the FTJ engine as a real drive letter (e.g. Z:\).

Standard Windows tools format NTFS, write files, and read data against our simulated FTJ memory.

Why IP Licensing? (Can We Flash NAND?)

The reality of semiconductor business: You cannot firmware-flash NAND into FTJ.

The NAND Reality

Can we flash FTJ controller firmware onto existing SSDs?

No. Existing SSDs use 3D NAND flash silicon. NAND physics physically requires high-voltage block erases.

Software cannot change quantum physical materials. Flashing byte-level FTJ logic onto NAND flash is impossible because the underlying silicon gates cannot perform in-place writes.

The Silicon IP Licensing Model

We operate like ARM or Rambus: Fabless Silicon IP.

- We do not build \$5B fabrication foundries.
- We license **Controller RTL blocks & FTL algorithms** to memory makers (Micron, Western Digital, SK Hynix, Kioxia).
- **Near-Term Hybrid SSDs:** 16GB of fast FTJ (as write buffer / KV-cache) + 4TB of cheap NAND. Our controller orchestrates both.

The 28nm Mature Node Strategic Advantage

Why high-margin deep tech does not require bleeding-edge sub-7nm EUV lithography.

Leading-Edge Trap (< 7nm)

- Requires rare ASML Extreme Ultraviolet (EUV) lithography.
- Wafer fabrication cost exceeds **\$20,000+ per wafer**.
- Dominated by TSMC; intense competition from Apple, Nvidia, and AMD for fabrication capacity.
- Tape-out mask set alone costs **\$5M - \$15M**.

The 28nm Mature Foundry Play

- **Abundant Global Capacity:** Fabricated at GlobalFoundries, UMC, TSMC, and SMIC.
- **Cost Advantage:** Wafer costs are ~**\$2,000** (10x cheaper).
- **BEOL Integration:** FTJ crossbars are deposited in Back-End-of-Line metal layers on top of standard 28nm CMOS transistors.
- **Sovereign Silicon:** Enables local grant funding (MeitY/PRAYAS) and domestic foundry tape-outs.

How to Talk to Engineers, Mentors & Investors

Word-for-word responses for high-stakes technical questions.

What did you actually build?

"I built an architectural simulator and Verilog RTL controller for Ferroelectric Tunnel Junction (FTJ) memory. It models crossbar physics like IR-drop and Merz's law, features lock-free NVMe queues, and mounts as a live Windows filesystem via WinFSP."

Why not just use DRAM?

"DRAM is volatile—if power cuts, all data is lost. It also constantly leaks power needing active refresh. FTJ provides near-DRAM speed with true non-volatile persistence and zero standby power."

Why FTJ over standard SSDs for AI?

"AI KV-cache workloads do heavy random writes. 3D NAND is block-based, so it suffers from 5x-10x write amplification and burns out its 3,000-cycle endurance in months. FTJ has $>10^{10}$ endurance and byte-level writes."

What is your go-to-market plan?

"Fabless Silicon IP licensing. We license the digital controller blocks and FTL algorithms to memory makers and enterprise FPGA accelerator vendors targeting hybrid SCM storage."